Foldable but not dockable: an over-squashing account of AlphaFold's protein–protein interface failures on PINDER

Graph Representation Learning — mini-project report.
Code: `https://github.com/muuki2/grl-oversquashing-pinder`

Abstract. AlphaFold folds individual chains confidently yet often mis-places the *complex*. We ask whether this is an *over-squashing* phenomenon: the interface is a sparse *inter*-chain bottleneck a message-passing network must route through, whereas folding each chain is bottleneck-free. We study AlphaFold's interface *sub-task*—predicting interface residues—with a GNN we own, on PINDER's leakage-free split, using predicted monomers (+pLDDT) as input and the bound complex only for labels (*no folding inference*). A synthetic probe establishes the mechanism cleanly (ρ=−0.97/+0.97 for resistance/Jacobian vs. accuracy). Our central real-data result is that over-squashing at the interface is **contingent on node-feature richness**: varying *only* the per-residue features from weak (degree+pLDDT, one-hot) to rich (ESM2), the benefit of cross-chain rewiring ΔAUPRC(knn−none) *flips sign* from +0.086 to −0.017—it relieves a genuine bottleneck when features are weak, but mildly *over-smooths* once rich features let the model solve the interface locally. In the weak regime the mechanism is causal: every rewiring beats the disconnected baseline (+0.03–0.04 AUPRC, ECE 0.084→0.038; a curvature-targeted SDRF matches them adding only ∼48 edges/system), and the trained-model Jacobian rises from 0 to $\sim 5\times10^{-2}$. Per-system failure tracks the *modelled* resistance ($\rho = -0.49$, confirmed within-system $\rho = -0.13$) but not the pose-unseen *native* one ($\rho \approx -0.11$)—a principled dissociation for *pose-free* inputs. We report bootstrap CIs, one-sided tests, and a minimum-detectable-effect throughout.

1 Introduction

AlphaFold(-Multimer) produces confidently folded monomers but a large fraction of its *complex* predictions are wrong—the documented "great monomers, weak complexes" regime (Bryant et al., 2022; Evans et al., 2022). Why should placing two well-folded chains relative to one another be so much harder than folding each? We propose a graph-theoretic reframing grounded in a core property of message-passing GNNs: *over-squashing*, the exponential compression of information forced through a graph bottleneck (Alon & Yahav, 2021; Topping et al., 2022; Di Giovanni et al., 2023). Modelled as a graph, a folded chain is internally dense, so intra-chain reasoning has *no* bottleneck; the interface between two chains is a thin neck of a few contacts through which all cross-chain signal must squeeze—precisely the topology in which message passing over-squashes.

Research question. *Is the difficulty of inferring a protein–protein interface governed by the graph bottleneck between the two chains (an over-squashing effect) rather than by how confidently each chain is folded, and does this survive on PINDER's leakage-free split?* This tests whether a *course concept*— information propagation / over-squashing—*quantitatively* explains a real failure mode, and whether the textbook *fix* (rewiring) transfers; we *measure* the relevant properties (effective resistance, Jacobian) rather than merely invoking them. Our answer is the main finding and it is *conditional*: the bottleneck binds—and rewiring helps—only when the features are too weak to solve the interface locally; with rich features the same cross-chain edges over-smooth.

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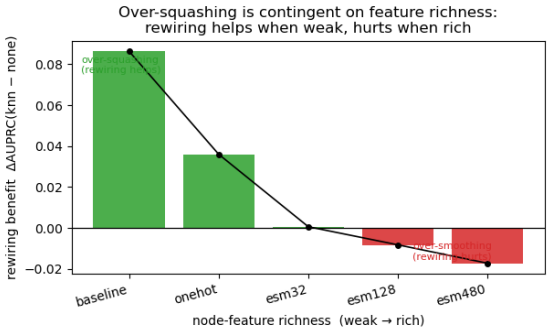
2 Methods and experimental setup

Design and diagnostics. We use a two-tier *synthetic probe* → *real validation* structure with the *same* two over-squashing diagnostics threading both tiers, so the real-data metric is calibrated against ground truth: the *effective resistance* $R_{\text{eff}}(u,v)=(\boldsymbol{e}_u-\boldsymbol{e}_v)^\top L^+(\boldsymbol{e}_u-\boldsymbol{e}_v)$ across the interface, the canonical bottleneck measure for over-squashing (Black et al., 2023); and the *Jacobian sensitivity* $\|\partial \boldsymbol{h}_v^{(T)}/\partial \boldsymbol{x}_u\|$ of a read-out node v to a source input u (Topping et al., 2022), measured both over freshly initialised models (an architecture property) and *across the interface on the trained real models*. Conceptually the bottleneck limits not just information flow but *representational capacity*: the cross-chain functions a fixed-width, fixed-depth MPNN can express degrade with the interface's width/depth/topology (Di Giovanni et al., 2023). The *synthetic probe* joins two Erdős–Rényi communities (16 nodes each) by exactly k cross-edges and reads a B-source label out at the A nodes, so the task *requires* cross-interface flow; we sweep $k \in \{1,2,4,8,16\}$ and depth $\in \{2,4\}$ (Supplementary Fig. 3).

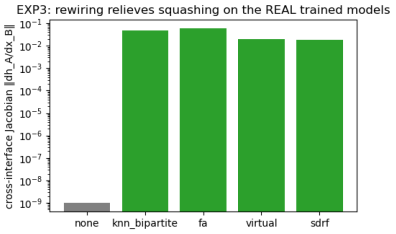
Real study (PINDER, inference-free). Per system we use the two AlphaFold-DB *predicted monomers* (pose-free; pLDDT in the B-factor column) as input and the *bound* complex only for labels. Each monomer is an intra-chain graph (residues within 8 Å of C_α); node features are per-residue **ESM2** embeddings (Lin et al., 2023), pLDDT, contact degree, and a chain indicator. The two monomers are joined by a *pose-free* cross-chain edge layer—the bottleneck knob: `none` (disconnected), `knn` (embedding-similarity bipartite edges), `fa` (fully-adjacent final layer), `virtual` (global hub), and `sdrf`, a *curvature-targeted* rewiring (Balanced-Forman, adding only the few edges that relieve the most negatively curved necks) (Topping et al., 2022). Interface labels (within 8 Å of the other chain in holo) are transferred onto predicted residues by *sequence alignment*. The predictor is a 4-layer GIN (64 hidden, LayerNorm + residuals), class-weighted CE, Adam (2×10^{-3}), 120 epochs; training is *deterministic* (TF32 off) and averaged over 10 seeds. We report AUPRC/AUROC/ECE and a base-rate-free *skill* = AUPRC − prevalence. We study heterodimers, skip degenerate systems (prevalence > 0.6 or < 0.02), and—being inference-free—test AlphaFold's interface *sub-task*, not end-to-end DockQ.

Two resistances and the richness sweep. The key design choice is *which* bottleneck to correlate with skill. Rather than the *native* resistance $R_{\text{eff}}^{\text{nat}}$ (holo contact graph, which the model never sees), we make the *modelled* resistance $R_{\text{eff}}^{\text{mod}}$ primary: the effective resistance *on the model's own pose-free input graph* between the true-interface residues, the bottleneck the network actually routes through ($R_{\text{eff}}^{\text{nat}}$ is kept as a contrast). We also run a high-power *within-system* test: per system we sweep the modelled width $k \in \{1,2,4,8\}$ and, centering both quantities within each system, correlate them—a paired design removing between-system noise. For the **headline experiment** we hold the GIN fixed and vary *only* the node features along a richness ladder—`baseline` (3-d), `onehot` (24-d), and ESM2 truncated to $\{32,128,480\}$-d—each regime *self-consistent* (its kNN layer rebuilt from its own features), reading off the rewiring benefit ΔAUPRC=AUPRC(`knn`)−AUPRC(`none`). Statistics use pre-registered *one-sided* Spearman tests, bootstrap CIs, and a minimum-detectable-effect 0.16 at n=243.

Hypotheses. **H0** (headline) the rewiring benefit depends on feature richness: ΔAUPRC > 0 for weak features, < 0 for rich ones (over-squashing → over-smoothing). **H1** skill ↓ as $R_{\text{eff}}^{\text{mod}}$ ↑, tested cross- and within-system. **H2** cross-chain rewiring recovers accuracy (*causal*), with the



(a) Rewiring benefit ΔAUPRC vs. feature richness: a sign-flip crossover.



(b) Cross-interface Jacobian on the trained models (weak regime).

Figure 1: Over-squashing on real data, and its limits. (a) The headline: the rewiring benefit flips from +0.086 to −0.017 AUPRC as features go weak→rich (over-squashing → over-smoothing). (b) In the weak regime the cross-interface gradient is 0 for the disconnected none (maximal squashing) and non-zero once cross edges carry interface signal—the mechanism behind Table 1.

trained-model Jacobian showing squashing is *relieved*. **H3** failures track *topology* more than AlphaFold's confidence (monomer pLDDT). **H4** flexibility (predicted→holo C_α-RMSD) and co-evolution (N_{eff}) are competing explanations. **H5** the link holds on the de-leaked *test* split.

3 Results

Synthetic probe: the mechanism is clean (H1/H2). As the bottleneck tightens ($k\downarrow$), A-node accuracy collapses toward chance, and the *same* two diagnostics we later carry to PINDER move in lockstep: across k, effective resistance anti-correlates with accuracy (ρ=−0.97) and the Jacobian correlates (ρ=+0.97). Depth does not rescue a tight bottleneck (the 2-layer curve never recovers; the 4-layer one needs $k \gtrsim 4$; Supplementary Fig. 3). This both establishes the mechanism and *calibrates* the metric.

Headline (H0): over-squashing at the interface is contingent on feature richness. The textbook prediction—rewiring relieves the bottleneck, so it helps—turns out to hold only in a specific regime. Holding the GIN fixed and varying *only* the per-residue features along the richness ladder (each regime self-consistent), the rewiring benefit ΔAUPRC(knn−none) *crosses zero* (Fig. 1a): +0.086 (degree+pLDDT), +0.036 (one-hot), +0.001 (ESM-32), −0.008 (ESM-128), −0.017 (ESM-480). With weak features the model cannot identify interface residues locally, so it *must* route signal across the sparse interface, and the bottleneck over-squashes it—cross-chain edges relieve this and help. With rich ESM features each residue already carries enough evolutionary and structural context to be classified locally, so the same edges no longer relieve a needed bottleneck and instead mildly *over-smooth* the two chains together, hurting slightly. The interface bottleneck is thus real but *conditional*—it binds only when the task forces cross-chain routing. This both sharpens the thesis and locates the regime for the causal tests below, which we therefore run in the weak (onehot) feature regime.

Real H2: in the weak regime, cross-chain rewiring is causal and relieves squashing. Table 1 (the weak onehot regime) shows that *every* cross-chain rewiring beats the bottlenecked

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Table 1: Real H2 (weak onehot regime). Interface predictor under five cross-chain edge layers (PINDER test, n=243; mean over 10 seeds; std ≤ 0.007 throughout). All cross-chain rewirings beat none; hub-free knn/fa/sdrf isolate bottleneck relief from the virtual node's added capacity.

rewiring	AUPRC	AUROC	ΔAUPRC	ECE
none	0.194	0.528	—	0.084
knn	0.232	0.589	+0.038	0.039
fa	0.224	0.566	+0.030	0.048
sdrf	0.230	0.584	+0.035	0.040
virtual	0.233	0.589	+0.039	0.038

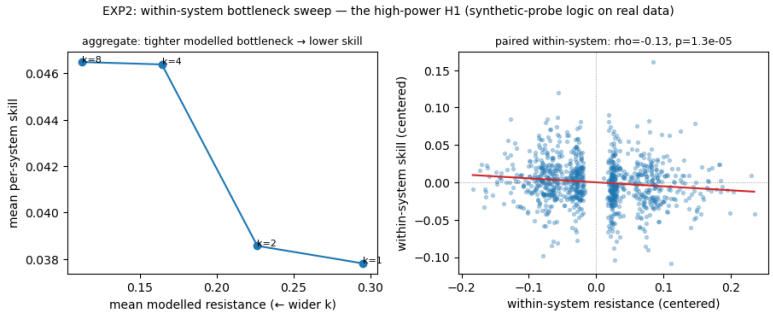


Figure 2: EXP2: within-system bottleneck sweep (the high-power H1). Left: aggregate per-system skill rises as the system's own modelled resistance falls (wider k). Right: centering both within each system gives a paired $\rho = -0.13$ (p=1.3×10^{-5}, 972 system–k points)—the synthetic probe's logic on real data. The cross-system H1 scatter and the H3/H4 partial-correlation bars are in the (excluded) supplementary.

none baseline (AUROC 0.528→0.57–0.59; +0.03–0.04 AUPRC; ECE more than halved, 0.084→0.038), over 10 deterministic seeds. knn, fa, and sdrf add cross edges *without* a global hub, so their gains isolate *bottleneck relief*, not the virtual node's added capacity; the curvature-targeted sdrf matches them adding only ~48 edges/system. The mechanism is confirmed directly on the trained models (Fig. 1b): the cross-interface Jacobian is identically 0 for the chain-disconnected none (maximal squashing) and rises to 4.8×10^{-2} (knn), 5.8×10^{-2} (fa) once cross edges carry signal—rewiring relieves squashing, not merely adds parameters.

Real H1 / EXP2: the *modelled* bottleneck predicts failure (SUPPORTED). Aligning the diagnostic with the graph the model actually uses turns the first iteration's null into a strong signal. Cross-system, per-system skill anti-correlates with the *modelled* resistance at $\rho = -0.49$ (one-sided p=2×10^{-16}, 95% CI $[-0.59, -0.39]$, sign-stable across all 10 seeds), whereas the *native* (pose-unseen) resistance stays weak ($\rho = -0.11$, p=0.075)—as it must, since the model never sees the holo geometry. A *within-system* sweep confirms it at high power: sweeping the modelled width k and centering within each system gives a paired $\rho = -0.13$ (p=1.3×10^{-5}, CI $[-0.19, -0.07]$, 972 system–k points), with skill falling as the system's own bottleneck tightens in 57% of systems (Fig. 2) —the synthetic probe's logic reproduced on real data.

Real H3/H4: topology beats confidence (SUPPORTED). In partial Spearman correlations (each term controlling for the others), the modelled-resistance term (ρ=−0.47, p=5.6×10^{-15}) is an order of magnitude larger than the strongest confidence term, monomer pLDDT ($|\rho|$=0.075),

and dwarfs flexibility and co-evolution N_{eff} (both $|\rho|<0.06$, $p>0.3$; Supplementary Fig. 4)—so H4's competitors are informative in the negative. One honest caveat: AlphaFold-Multimer's *complex* confidence (ipTM/PAE), the fairer contest, was not exposed by PINDER, so our baseline is the weaker *monomer* pLDDT; at n=243 the minimum detectable $|\rho|$ is 0.16, so these small terms are read as bounded, not absent.

Real H5: generalisation gradient (SUPPORTED). Stratifying *test* skill by *modelled*-resistance tertile (low→high) gives a monotone decline ($0.073 \to 0.035 \to 0.025$); we count H5 as supported because the powered continuous within-split H1 is itself significant, tying over-squashing to OOD generalisation. The large train→test skill drop reflects expected overfitting of a small GIN on 383 systems.

4 Relation to the literature and course concepts

This study operationalises several course concepts. *Over-squashing* and *information propagation* (Alon & Yahav, 2021) are measured directly via *effective resistance* (Black et al., 2023) and Jacobian sensitivity (Topping et al., 2022; Di Giovanni et al., 2023); *graph rewiring* as an inductive-bias intervention is the remedy literature (virtual nodes, fully-adjacent layers, curvature/spectral rewiring—SDRF (Topping et al., 2022), BORF (Nguyen et al., 2023), FOSR (Karhadkar et al., 2023)—and expander propagation, EGP (Deac et al., 2022)), of which our H2/SDRF is a domain instance, while the *expressive-power* view (Di Giovanni et al., 2023) explains why depth alone cannot rescue a tight neck. Our crossover places *over-squashing* and *over-smoothing* (NT & Maehara, 2019; Rusch et al., 2023) on one feature-richness axis. Finally, PINDER's *leakage-free* split (Kovtun et al., 2024) frames the question as *generalisation under shift*, and we reframe the AlphaFold-Multimer "confident monomers, wrong complex" failure (Evans et al., 2022; Bryant et al., 2022) (cf. interface predictors like MaSIF (Gainza et al., 2020)) as a GNN information-flow problem.

5 Discussion, critique, and outlook

The central lesson is that over-squashing at the interface is *real but contingent*, and the crossover maps when it binds: weak features force cross-chain routing and *over-squashing* dominates (rewiring helps); rich ESM features make the interface locally solvable and the same edges *over-smooth* (rewiring hurts). The textbook fix is therefore regime-dependent—more honest than "rewiring always helps," and a caution against rewiring blindly when features are already expressive. In the weak regime the evidence is coherent (causal: hub-free rewirings and a parsimonious SDRF rule out pure capacity, and the Jacobian shows squashing relieved; observational: the *modelled* bottleneck tracks failure at $\rho=-0.49$); the one dissociation—the pose-unseen *native* resistance is weak—is principled for *pose-free* inputs.

We are candid about limits: the absolute predictor is weak (test AUROC 0.589, skill $\approx+0.04$), so claims rest on *relative* effects exceeding their 10-seed std, not a strong classifier; weak correlations are reported with bootstrap CIs, one-sided tests, and an MDE so they read as bounded, not headline. Future work should use *pose-aware* inputs, move to *edge-level* contact prediction, and—given a per-system AF2-Multimer DockQ table—test the thesis on AlphaFold's true output rather than its sub-task.

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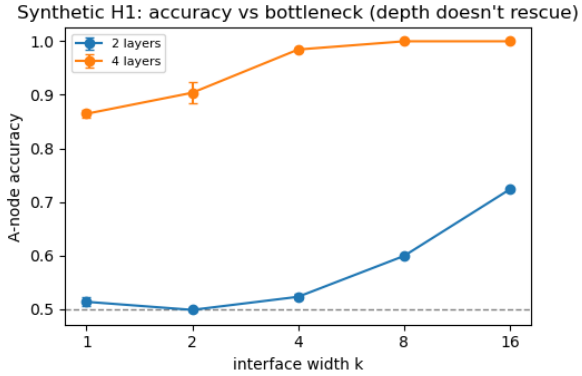
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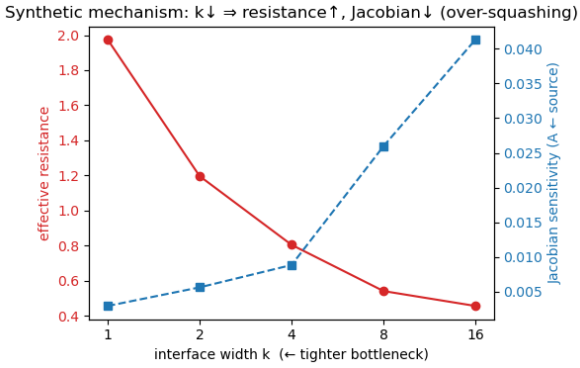
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Supplementary material

A. Synthetic probe. Two Erdős–Rényi communities (A, B; 16 nodes each) joined by exactly k cross-edges; a single B-source label is read out at the A nodes, so the task requires cross-interface flow. Figure 3 reports A-node accuracy, cross-interface R_{eff}, and Jacobian sensitivity across $k \in \{1, 2, 4, 8, 16\}$ and depth $\in \{2, 4\}$.

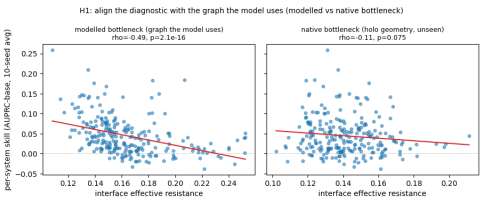


(a) Accuracy vs. bottleneck width k.

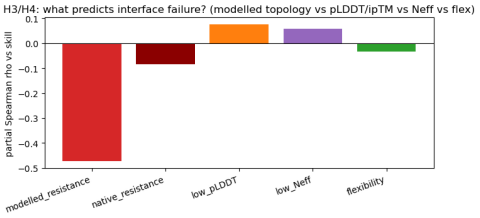


(b) Resistance ↑, Jacobian ↓ as k↓.

Figure 3: Synthetic probe. A controllable interface bottleneck reproduces over-squashing: accuracy collapses as k tightens, quantitatively tracked by effective resistance and Jacobian sensitivity (ρ=−0.97/+0.97 vs. accuracy)—the metrics we carry to PINDER.



(a) H1: modelled (left) vs. native (right) resistance vs. skill, cross-system.



(b) H3/H4: partial-correlation predictors of failure.

Figure 4: Cross-system predictors. (a) The *modelled* bottleneck out-tracks the pose-unseen *native* one. (b) Modelled topology dwarfs confidence (pLDDT), flexibility, and N_{eff}.

B. Cross-system H1 and H3/H4 predictors.

C. Hypothesis scorecard. Table 2 collects the predicted-vs-observed verdict for every hypothesis (each also stated in the running text of §3).

D. Implementation. GIN: 4 layers, 64 hidden, LayerNorm + residuals; class-weighted cross-entropy (positive weight capped at 3); Adam (2×10^{-3}, weight decay 10^{-5}), 120 epochs; deterministic training (TF32 off, deterministic algorithms), 10 seeds. ESM2 features from `esm2_t12_35M_UR50D` (480-d), cached per system. Full code, the executed notebook, and the result tables/CSVs are in the repository linked above.

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Table 2: Hypothesis scorecard. Predicted vs. observed (PINDER test, n=243, 10 seeds).

	prediction	observed	verdict
H0	rewiring benefit depends on features	ΔAUPRC flips $+0.086 \rightarrow -0.017$	**supported (crossover)**
H1	skill $\downarrow$ as $R_{\text{eff}}^{\text{mod}}$ $\uparrow$	$\rho = -0.49$ cross-sys; -0.13 within	**supported**
H2	rewiring recovers accuracy	all knobs >none; Jacobian relieved	**supported (causal)**
H3	topology > confidence	modelled -0.47 vs. pLDDT 0.075	**supported**
H4	flexibility/N_{eff} compete	neither predicts failure	negative (informative)
H5	link holds on test split	monotone test tertiles, gated on H1	**supported**

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